

SCENARIODIFF: A Scenario-level Guidance Framework for Multimodal Time Series Forecasting

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Abstract—Textual context such as news, reports, and logs can provide valuable signals for time series forecasting, especially when future dynamics are driven by external events that are not yet visible in historical values. Existing multimodal forecasting methods often either ask large language models (LLMs) to predict numerical values directly or fuse text and time series implicitly, making contextual influence difficult to interpret and control. We propose SCENARIODIFF, a hierarchical contextual reasoning framework for multimodal time series forecasting under noisy and weakly aligned documents. SCENARIODIFF organizes contextual information into three levels: a *Historical Context Agent* extracts stepwise evidence from raw documents, a *Scenario Agent* produces a qualitative scenario description for the forecast horizon, and an *Anchor Guidance Agent* generates sparse anchor points for event-relevant future regions. These structured signals condition a *Multimodal Diffusion Transformer*, while *Anchor Blended Sampling* locally refines generated trajectories without retraining. Experiments on the Time-MMD benchmark show that SCENARIODIFF is especially effective in event-driven domains, demonstrating the value of explicit hierarchical scenario guidance for multimodal time series forecasting. Our full implementation is available at <https://github.com/ttb06/ScenarioDiff>.

Index Terms—time series forecasting, multimodal forecasting, diffusion models, large language models, scenario guidance

I. INTRODUCTION

Time series forecasting (TSF) supports decision-making in domains such as energy [1], economics [2], healthcare [3], and transportation [4]. Despite its importance, most forecasting models operate primarily on numerical histories. In practice, however, human experts rarely interpret temporal signals in isolation. An energy engineer may analyze demand traces together with weather forecasts and maintenance logs, while a financial analyst may read price series alongside earnings reports and macroeconomic news. Such contextual information can explain historical dynamics and, more importantly, indicate future developments that are not yet visible from numerical values alone. Figure 1 illustrates how contextual

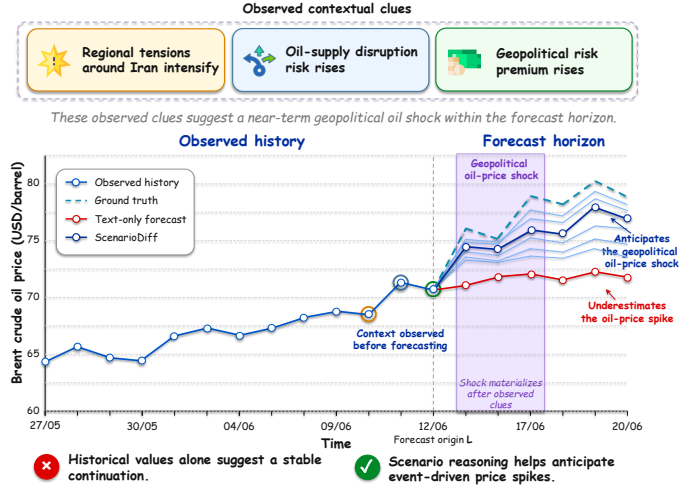


Fig. 1: **Motivating example.** Brent crude oil prices around the Israel–Iran conflict in June 2025 illustrate the need for scenario-level reasoning. Before the forecast origin L , historical prices show only a moderate upward trend, while available contextual clues indicate rising regional tensions, oil-supply disruption risk, and geopolitical risk premium. These signals suggest a near-term oil-price shock that is not fully reflected in the numerical history. SCENARIODIFF uses the observed context to form a disruption scenario and produces trajectories that better follow the realized price increase.

evidence provides useful clues about future dynamics beyond what is visible from the numerical history alone.

This observation has motivated growing interest in *Multimodal Time Series Forecasting* (MTSF), where numerical histories are paired with contextual text such as news, reports, and logs. Existing studies show that textual information can improve forecasting across domains [5]–[8]. However, real-world text is often noisy, redundant, weakly aligned with timestamps, and only partially relevant to the target series. Under such conditions, implicit text–time fusion makes it difficult to identify which evidence influences the forecast and how strongly the model should trust it.

Large language models (LLMs) are a natural tool for

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processing unstructured context because they can summarize, filter, and reason over textual evidence. Recent studies have explored two main directions. The first direction prompts or reprograms numerical histories into language-like inputs and asks LLMs to predict future values directly [9]. While appealing, this places long-horizon numerical generation on models primarily optimized for language understanding rather than precise temporal extrapolation [10]. The second direction incorporates text through multimodal fusion, cross-attention, or agent-based summarization pipelines [5]–[8]. These methods are effective, but often assume clean text–timestamp alignment or absorb contextual evidence implicitly in latent space.

An additional challenge is that contextual text may describe not only smooth trend continuation, but also event-driven deviations such as policy interventions, outages, or abrupt demand shifts. Capturing such cases requires future-oriented hypotheses that can guide a numerical forecaster without asking an LLM to directly output numbers. It also requires a mechanism for translating coarse textual expectations into localized constraints when abrupt changes are likely.

In this work, we propose SCENARIODIFF, a hierarchical contextual reasoning framework for MTSF. The key idea is to organize contextual information into three complementary levels of guidance, each serving a different role in guiding the forecast. First, a frozen *Historical Context Agent* compresses raw documents into *stepwise context summaries* aligned with the observed history, grounding the model in document-level evidence. Second, a frozen *Scenario Agent* synthesizes the historical series and these summaries into a *scenario description*, which serves as a qualitative prior over the forecast horizon. Third, an *Anchor Guidance Agent* converts the available context into sparse *anchor points*, providing time-localized guidance for future regions where abrupt changes are likely. This hierarchy separates past evidence, future hypotheses, and local trajectory constraints, making contextual influence more explicit and modular than implicit fusion.

We instantiate the forecaster as a *Multimodal Diffusion Transformer*, which models future trajectories through iterative denoising conditioned on the structured signals produced by the agents. The stepwise context summaries and scenario description guide the base diffusion process, allowing the model to combine numerical history with scenario-level textual evidence without asking LLMs to directly output numerical forecasts. To make anchor points actionable, we introduce *Anchor Blended Sampling*, an inference-time refinement procedure that locally edits anchor-relevant regions using a distance-to-band objective while preserving non-anchor regions through blended diffusion. Thus, anchor points act as soft local constraints rather than hard overrides, encouraging event-consistent trajectories without retraining the base forecaster.

We evaluate SCENARIODIFF on the **Time-MMD** benchmark [5]. Results show that SCENARIODIFF achieves strong performance, with the clearest gains in event-driven domains where textual evidence contains actionable signals about future dynamics. This supports our central hypothesis that explicit hierarchical scenario guidance is most beneficial when future

changes are driven by external events weakly reflected in the numerical history.

In summary, our main contributions are as follows: (i) We introduce SCENARIODIFF, a hierarchical contextual reasoning framework that transforms noisy textual documents into three levels of guidance for MTSF: historical evidence, future scenario, and local anchor constraints. (ii) We integrate these structured signals into a Multimodal Diffusion Transformer, enabling probabilistic forecasting conditioned on explicit historical and scenario-level context. (iii) We propose Anchor Blended Sampling, which connects anchor points from the Anchor Guidance Agent with local post-hoc trajectory refinement without retraining the diffusion forecaster. (iv) We provide empirical analysis on **Time-MMD**, showing that the main gains concentrate in event-driven domains where textual evidence carries actionable future signals.

II. RELATED WORK

LLM-Based Time Series Forecasting. Recent work has explored large language models (LLMs) as sequence priors for time series analysis and forecasting. OFA [11] studies how pre-trained language-model backbones can be transferred to time series tasks. Time-LLM [9] reprograms numerical histories into language-like representations, enabling frozen LLMs to perform few-shot and zero-shot forecasting. CALF [12] further improves this paradigm by aligning temporal tokens with textual representations. These methods demonstrate the potential of LLMs to provide semantic priors for forecasting. However, they often require LLMs to support numerical extrapolation, which is not ideal for precise long-horizon prediction due to the mismatch between language modeling and fine-grained temporal generation. In contrast, SCENARIODIFF uses LLMs for producing contextual information rather than direct numerical generation: the agents summarize historical evidence, produce scenario descriptions, and extract anchor points, while a diffusion forecaster handles probabilistic generation.

Multimodal Time Series Forecasting. Multimodal time series forecasting incorporates contextual text such as news, reports, logs, and event descriptions into numerical forecasting models. Time-MMD [5] introduces a multi-domain benchmark and shows that paired textual information can improve forecasting across domains. TaTS [6] aligns text with each timestep and uses text embeddings as additional temporal features. TimeCAP [13] adopts an agent-based pipeline to summarize and contextualize time series before prediction, while ChatTime [7] jointly models numerical and textual inputs in a unified Transformer. Although effective, these approaches often assume clean text–timestamp alignment or fuse text implicitly in latent space, making it difficult to inspect which evidence affects the forecast under noisy, weakly aligned, multi-document inputs. SCENARIODIFF differs by converting raw documents into explicit stepwise context summaries, a scenario description, and anchor points, making the contextual roles more modular and interpretable.

Diffusion Models for Time Series Forecasting. Diffusion models have become a strong class of probabilistic forecasters,

generating future trajectories through iterative denoising. Existing methods include one-shot horizon generation [14], [15] and autoregressive diffusion-based prediction [16]. Subsequent work improves long-horizon forecasting with multi-resolution denoising, structured backbones, and retrieval-augmented conditioning [17]–[19]. Recent multimodal diffusion methods further inject timestamps and textual context through fusion, cross-attention, or classifier-free guidance [20], [21], while flow-matching models aim to reduce sampling cost [22], [23]. Most prior methods for condition generation are mainly through architecture-level fusion and do not explicitly control how external context shapes the sampling. Our work builds on diffusion forecasting but differs in two aspects. First, the conditioning signals are not raw text or generic latent embeddings; instead, they are structured scenario-level signals that separately encode historical evidence and future hypotheses. Second, beyond architectural conditioning, we introduce an inference-time guidance mechanism that steers generated trajectories toward sparse anchor bands implied by contextual evidence. This provides an interpretable and flexible way to shape the forecasting distribution while remaining compatible with standard diffusion backbones.

III. PRELIMINARIES

A. Denoising Diffusion Probabilistic Models

Denoising Diffusion Probabilistic Models (DDPMs) [24] are generative models that iteratively add Gaussian noise to a clean sample, $\mathbf{x}^{(0)}$, over T steps, following a Markov chain. The noise schedule, $\{\beta_t\}_{t=1}^T$, consists of $\beta_t \in (0, 1)$, and the cumulative noise is represented by $\bar{\alpha}_t = \prod_{i=1}^t (1 - \beta_i)$. The forward diffusion process is described by:

$$q(\mathbf{x}^{(1:T)} | \mathbf{x}^{(0)}) = \prod_{t=1}^T q(\mathbf{x}^{(t)} | \mathbf{x}^{(t-1)}), \quad (1)$$

$$q(\mathbf{x}^{(t)} | \mathbf{x}^{(t-1)}) = \mathcal{N}(\sqrt{1 - \beta_t} \mathbf{x}^{(t-1)}, \beta_t \mathbf{I}). \quad (2)$$

This process allows us to directly sample $\mathbf{x}^{(t)}$ from the clean sample $\mathbf{x}^{(0)}$ as:

$$\mathbf{x}^{(t)} = \sqrt{\bar{\alpha}_t} \mathbf{x}^{(0)} + \sqrt{1 - \bar{\alpha}_t} \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}). \quad (3)$$

To reverse this diffusion chain, DDPM learns to progressively remove noise in each step. The reverse process is also modeled as a Markov chain and is defined as:

$$p_\theta(\mathbf{x}^{(0:T)}) = p_\theta(\mathbf{x}^{(T)}) \prod_{t=1}^T p_\theta(\mathbf{x}^{(t-1)} | \mathbf{x}^{(t)}), \quad (4)$$

$$p_\theta(\mathbf{x}^{(t-1)} | \mathbf{x}^{(t)}) = \mathcal{N}(\mu(\mathbf{x}^{(t)}, \mathbf{x}_\theta(\mathbf{x}^{(t)}, t)), \Sigma_t). \quad (5)$$

Here, $p_\theta(\mathbf{x}^{(T)})$ is chosen as a standard Gaussian $\mathcal{N}(\mathbf{0}, \mathbf{I})$, while $\mathbf{x}_\theta(\mathbf{x}^{(t)}, t)$ is the predicted clean sample $\mathbf{x}^{(0)}$ from the noisy input $\mathbf{x}^{(t)}$, and Σ_t is typically a fixed variance function.

B. Conditional DDPM For Time Series Forecasting

In time series forecasting, we extend the DDPM framework to handle conditional inputs. Let $\mathbf{c}$ represent the conditioning signals; the goal of the denoising network is to predict the

future horizon, $\mathbf{x}^{(0)} \equiv \mathbf{x}_{L+1:L+H} = \langle \mathbf{x}_{L+1}, \mathbf{x}_{L+2}, \dots, \mathbf{x}_{L+H} \rangle$, based on the noisy future time series $\mathbf{x}^{(t)}$ and the condition $\mathbf{c}$. The denoising objective is formulated as:

$$\mathcal{L}(\theta) = \mathbb{E}_{\mathbf{x}^{(0)}, t, \boldsymbol{\epsilon}} \left[\|\mathbf{x}^{(0)} - \mathbf{x}_\theta(\mathbf{x}^{(t)}, t, \mathbf{c})\|_2^2 \right]. \quad (6)$$

Normally, $\mathbf{c}$ could be the lookback window $\mathbf{x}_{1:L}$. However, in our setting, we also integrate textual context to enrich the condition. Specifically, for each historical timestep $n \in \{1, \dots, L\}$, there is a local bag $\mathcal{M}_n = \{s_i\}_{i=1}^{|\mathcal{M}_n|}$, which contains textual descriptions of events occurring at that point.

IV. METHODOLOGY

A. Framework Overview

Fig. 2 summarizes SCENARIODIFF. Let $\mathbf{x}_{1:L} \in \mathbb{R}^L$ denote the observed lookback window and $\mathbf{x}_{L+1:L+H} \in \mathbb{R}^H$ denote the future horizon. For each historical timestep $n \in \{1, \dots, L\}$, we assume access to an aligned document set $\mathcal{M}_n$. SCENARIODIFF organizes contextual reasoning into three hierarchical levels. First, a frozen *Historical Context Agent* maps $\mathcal{M}_{1:L}$ to *stepwise context summaries* $\mathbf{s}_{1:L}^{\text{ctx}}$, grounding the model in document-aligned historical evidence. Second, a frozen *Scenario Agent* maps $(\mathbf{x}_{1:L}, \mathbf{s}_{1:L}^{\text{ctx}})$ to a *scenario description* $\mathbf{s}^{\text{scn}}$, projecting a coarse qualitative prior over the forecast horizon. Third, an *Anchor Guidance Agent* maps $(\mathbf{x}_{1:L}, \mathbf{s}_{1:L}^{\text{ctx}}, \mathbf{s}^{\text{scn}})$ to sparse *anchor points* $\mathcal{A}$, which provide localized constraints for inference-time trajectory refinement.

We instantiate the numerical forecaster as a conditional denoising diffusion model that estimates $p(\mathbf{x}_{L+1:L+H} | \mathbf{x}_{1:L}, \mathbf{s}_{1:L}^{\text{ctx}}, \mathbf{s}^{\text{scn}})$. Let $\mathbf{x}^{(0)} \equiv \mathbf{x}_{L+1:L+H}$ be the clean future target. The forward process corrupts it with Gaussian noise: $q(\mathbf{x}^{(t)} | \mathbf{x}^{(0)}) = \mathcal{N}(\sqrt{\bar{\alpha}_t} \mathbf{x}^{(0)}, (1 - \bar{\alpha}_t) \mathbf{I})$, or equivalently, $\mathbf{x}^{(t)} = \sqrt{\bar{\alpha}_t} \mathbf{x}^{(0)} + \sqrt{1 - \bar{\alpha}_t} \boldsymbol{\epsilon}$, $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$. During denoising, the model receives the observed history $\mathbf{x}_{1:L}$, the noisy future $\mathbf{x}^{(t)}$, and the contextual conditions produced by the Historical Context Agent and the Scenario Agent. Starting from Gaussian noise, the reverse process generates a trajectory consistent with both numerical history and scenario-level context. Anchor Blended Sampling then uses $\mathcal{A}$ to locally refine the trajectory.

B. Hierarchical Contextual Reasoning

a) *Historical Context Agent*: Raw documents aligned with each historical timestep are often verbose, redundant, and only weakly related to the forecasting target. Directly encoding such documents may obscure useful event signals. We therefore use a frozen *Historical Context Agent* to compress each local document set into a concise summary.

For each timestep n , the agent takes $\mathcal{M}_n$ as input and produces a short summary s_n^{ctx} . If no useful evidence is available, the agent emits a special [NO_EVENT] token. Collecting these outputs yields the sequence $\mathbf{s}_{1:L}^{\text{ctx}} = \langle s_1^{\text{ctx}}, \dots, s_L^{\text{ctx}} \rangle$. These summaries provide a temporally grounded semantic view of the observed history and serve as the main textual representation of past evidence.

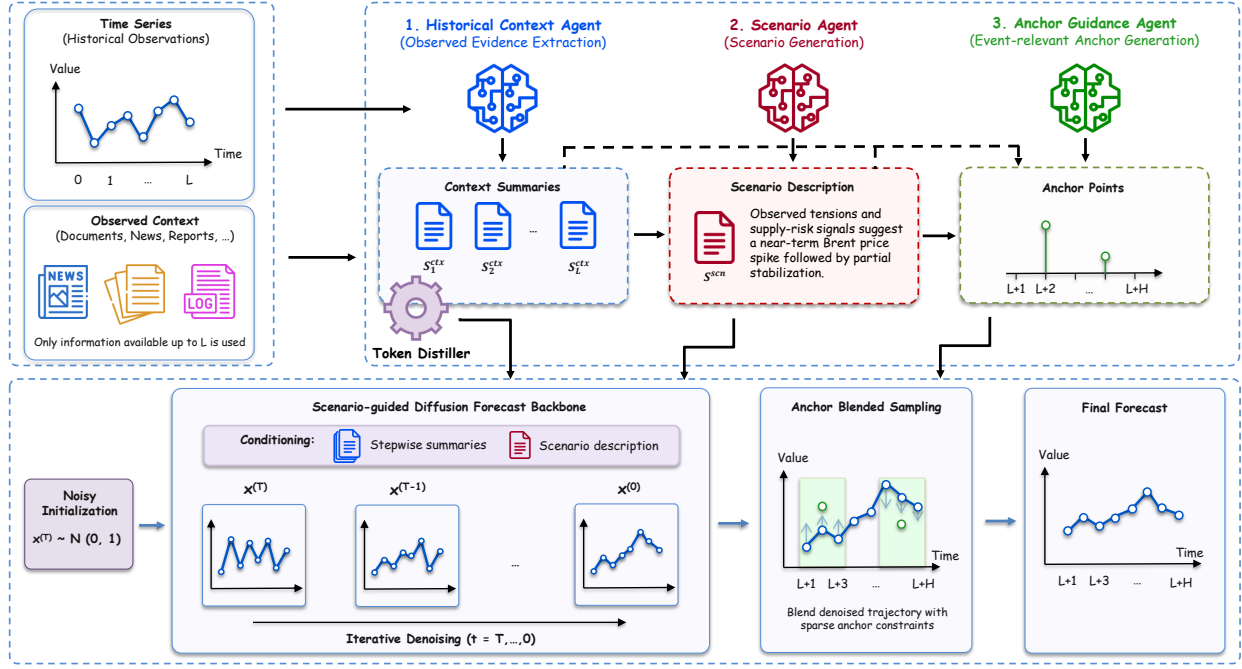


Fig. 2: SCENARIODIFF overview.

To obtain continuous conditioning features, each summary is encoded by a text encoder $\text{Enc}_{\text{text}}(\cdot)$ followed by an average pooling as:

$$\mathbf{c}_n^{\text{hist}} = \text{AvgPool}(\text{Enc}_{\text{text}}(\mathbf{s}_n^{\text{ctx}})), \quad \mathbf{c}_n^{\text{hist}} \in \mathbb{R}^{d_{\text{emb}}}. \quad (7)$$

We denote the encoded historical context sequence as $\mathbf{C}_{1:L}^{\text{hist}} = \langle \mathbf{c}_1^{\text{hist}}, \dots, \mathbf{c}_L^{\text{hist}} \rangle$. The resulting sequence $\mathbf{C}_{1:L}^{\text{hist}}$ is later injected into the historical tokens of the diffusion forecaster.

b) *Scenario Agent*: While the stepwise context summaries capture fine-grained historical evidence, forecasting also benefits from a global qualitative hypothesis about the future horizon. We introduce a frozen *Scenario Agent*, which generates a *scenario description* from the observed history and the stepwise context summaries: $\mathbf{s}^{\text{scn}} = \text{LLM}(\mathbf{x}_{1:L}, \mathbf{s}_{1:L}^{\text{ctx}})$. Because the same history-only procedure is used in both training and inference, this design avoids temporal leakage.

The scenario description is a short natural-language statement about expected future behavior, such as overall direction, volatility, or likely disruptions. We encode it with the same text encoder $\text{Enc}_{\text{text}}(\cdot)$:

$$\mathbf{C}^{\text{scn}} = \text{Enc}_{\text{text}}(\mathbf{s}^{\text{scn}}), \quad \mathbf{C}^{\text{scn}} \in \mathbb{R}^{N_f \times d_{\text{emb}}}. \quad (8)$$

This representation serves as a horizon-level prior for the diffusion forecaster.

c) *Scenario Consistency Score*: For analysis only, we define a Scenario Consistency Score (SCS) to measure whether the generated scenario is semantically close to what actually occurs. Let $\hat{\mathbf{s}}_{\text{oracle}}^{\text{scn}}$ be an oracle scenario description obtained by summarizing the ground-truth future $\mathbf{x}_{L+1:L+H}$ with the same prompt. This oracle is never used during training or inference. We compute SCS using a separate sentence-level diagnostic encoder $\text{Enc}_{\text{diag}}(\cdot)$:

$$\text{SCS}(\mathbf{s}^{\text{scn}}) = \cos(\text{Enc}_{\text{diag}}(\mathbf{s}^{\text{scn}}), \text{Enc}_{\text{diag}}(\hat{\mathbf{s}}_{\text{oracle}}^{\text{scn}})). \quad (9)$$

A higher SCS indicates stronger semantic agreement between the generated scenario and the realized future. We use SCS alongside distributional metrics (MMD, Fréchet distance) in our empirical analysis to verify that generated scenarios carry meaningful signal above noise baselines (Section V-C).

d) *Anchor Guidance Agent*: The scenario description provides a coarse future prior, but it does not directly specify where abrupt changes may occur. We therefore introduce an *Anchor Guidance Agent*, which converts the available context into sparse *anchor points* for inference-time guidance.

Given $\mathbf{x}_{1:L}$, $\mathbf{s}_{1:L}^{\text{ctx}}$, and $\mathbf{s}^{\text{scn}}$, the Anchor Guidance Agent outputs a set of M anchor points $\mathcal{A} = \{(t_j, l_j, u_j, w_j)\}_{j=1}^M$, where $t_j \in \{L+1, \dots, L+H\}$ is a future timestep, $[l_j, u_j]$ is a plausible value interval, and w_j is a confidence weight. We define

$$w_j = \text{clip}\left(c \left(1 + \frac{1}{\varepsilon + |u_j - l_j|}\right), 0, w_{\max}\right), \quad (10)$$

so that tighter intervals yield stronger guidance. These anchor points provide time-localized structure that complements the global scenario description and are used during inference to refine sampled trajectories.

C. Multimodal Diffusion Transformer

a) *Architecture*: We choose a Transformer-based denoising architecture that enhances the model's ability to capture the conditioning signal for time series. Given an input time series, we first apply RevIN [25] to reduce temporal distribution shift and outlier effects.

The normalized series is then transformed into tokens by a tokenization module $\Phi_{\text{ts}}(\cdot)$, which incorporates the diffusion step embedding t : $\mathbf{Z}_0 = \Phi_{\text{ts}}(\text{RevIN}([\mathbf{x}_{1:L}; \mathbf{x}_{L+1:L+H}^{(t)}], t)$.

Algorithm 1 Asymmetric conditioning in SCENARIODIFF

Require: $\mathbf{x}_{1:L}, \mathbf{x}_{L+1:L+H}, \mathbf{s}_{1:L}^{\text{ctx}}, \mathbf{s}_{1:L}^{\text{scn}}$, step t
Ensure: $\hat{\mathbf{x}}_{L+1:L+H}^{(0)}$
1: **Timestamp tokenization:**
2: $\mathbf{Z}_0 \leftarrow \Phi_{\text{ts}}(\text{RevIN}([\mathbf{x}_{1:L}; \mathbf{x}_{L+1:L+H}]), t)$
3: **Text encoding:**
4: $\mathbf{C}_h \leftarrow \text{Distill}(\text{Enc}_{\text{text}}(\mathbf{s}_{1:L}^{\text{ctx}}))$
5: $\mathbf{C}_s \leftarrow \text{Distill}(\text{Enc}_{\text{text}}(\mathbf{s}_{1:L}^{\text{scn}}))$
6: **for** $\ell = 0, \dots, D-1$ **do**
7: Split tokens:
8: $\mathbf{H}_\ell \leftarrow \mathbf{Z}_{\ell, 1:L}, \quad \mathbf{F}_\ell \leftarrow \mathbf{Z}_{\ell, L+1:L+H}$
9: Inject historical context:
10: $\tilde{\mathbf{H}}_\ell \leftarrow \mathbf{H}_\ell + \psi_h(\mathbf{C}_h)$
11: Inject scenario guidance:
12: $\tilde{\mathbf{F}}_\ell \leftarrow \mathbf{F}_\ell + \text{CA}(\mathbf{F}_\ell, \mathbf{C}_s, \mathbf{C}_s)$
13: Joint denoising update:
14: $\mathbf{Z}_{\ell+1} \leftarrow \mathcal{B}_\ell([\tilde{\mathbf{H}}_\ell; \tilde{\mathbf{F}}_\ell], t)$
15: **end for**
16: **Future reconstruction:**
17: $\hat{\mathbf{x}}_{L+1:L+H}^{(0)} \leftarrow \text{RevIN}^{-1}(\text{Head}(\mathbf{Z}_{D, L+1:L+H}))$
18: **return** $\hat{\mathbf{x}}_{L+1:L+H}^{(0)}$

b) Token Distiller: The textual conditions can be long, especially when many historical summaries are available. To reduce computation, we compress long contextual token sequences $\mathbf{C}^{\text{hist}}$ and $\mathbf{C}^{\text{scn}}$ into a compact window of Q distilled tokens, where $Q \ll L$. We introduce learnable prototype queries $\mathbf{R} \in \mathbb{R}^{Q \times D}$ and obtain the distilled context through cross-attention:

$$\mathbf{C}_{\text{distill}}^{\text{ctx}} = \text{CrossAttn}(\mathbf{R}, \mathbf{C}^{\text{ctx}}) \in \mathbb{R}^{Q \times D}, \quad (11)$$

$$\mathbf{C}_{\text{distill}}^{\text{scn}} = \text{CrossAttn}(\mathbf{R}, \mathbf{C}^{\text{scn}}) \in \mathbb{R}^{Q \times D}. \quad (12)$$

The learnable queries $\mathbf{R}$ aggregate salient information into fixed-size representations for downstream conditioning. Padding tokens are masked, and fully masked sequences produce zero outputs for stability. The distilled tokens condition the Transformer decoder layers through a cross-attention module, reducing the cost of processing long contexts.

c) Condition Injection: To inject conditional information during reverse denoising, the model uses adaptive layer normalization (AdaLN) [26]. Specifically, the diffusion-step embedding and the distilled contextual representations generate scale and shift parameters for Transformer hidden states. This allows the denoising network to adapt its computation at each diffusion step according to both temporal noise level and multimodal context.

The detailed architecture is summarized in Alg. 1. Here, Φ_{ts} denotes time-series tokenization, ψ_h is the additive history-side injection MLP, CA denotes cross-attention, $\mathcal{B}_\ell$ denotes an AdaLN Transformer block, and Distill compresses long contextual token sequences using learnable prototype queries.

Finally, the resulting patch tokens are processed by a stack of Transformer encoder layers with self-attention and feed-forward blocks, and a lightweight flatten-linear decoder maps the final hidden representations back to the reconstructed denoised time series.

Algorithm 2 Anchor Blended Sampling

Require: diffusion model $\mathbf{x}_\theta$, anchors $\mathcal{A}$, mask $\mathbf{m}$, edit steps T_e , scales $\{\gamma_t\}_{t=1}^{T_e}$
Ensure: refined sample $\mathbf{x}^{(0)}$
1: Sample $\mathbf{x}^{(T)} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
2: **for** $t = T, \dots, 1$ **do**
3: Draw $\mathbf{x}^{(t-1)}$ using the reverse diffusion kernel.
4: **end for**
5: Set source trajectory $\mathbf{x}_\theta^{(0)} \leftarrow \mathbf{x}^{(0)}$
6: Noise the source to level T_e : $\mathbf{x}_\theta^{(T_e)} \sim q(\mathbf{x}^{(T_e)} | \mathbf{x}_\theta^{(0)})$
7: **for** $t = T_e, \dots, 1$ **do**
8: Estimate posterior parameters μ_t, Σ_t
9: $g_t \leftarrow -\nabla_{\mathbf{x}^{(t)}} \mathcal{L}_{\text{anch}}(\mathbf{x}_\theta(\mathbf{x}^{(t)}), t; \mathcal{A})$
10: $\tilde{\mathbf{x}}^{(t-1)} \sim \mathcal{N}(\mu_t + \gamma_t \Sigma_t g_t, \Sigma_t)$
11: Draw $\mathbf{x}_\theta^{(t-1)} \sim q(\mathbf{x}^{(t-1)} | \mathbf{x}_\theta^{(0)})$
12: $\mathbf{x}^{(t-1)} \leftarrow \mathbf{m} \odot \tilde{\mathbf{x}}^{(t-1)} + (1 - \mathbf{m}) \odot \mathbf{x}_\theta^{(t-1)}$
13: **end for**
14: **return** $\mathbf{x}^{(0)}$

D. Anchor Blended Sampling

After standard reverse diffusion, we obtain an initial forecast $\mathbf{x}_\theta^{(0)}$, which is used as the source trajectory for a post-hoc editing stage. Anchor Blended Sampling refines this source trajectory by editing anchor-relevant regions while preserving the remaining temporal structure. Unlike image-based blended diffusion, which often uses CLIP guidance [27], our refinement is defined by the distance-to-band objective $\mathcal{L}_{\text{anch}}(\mathbf{x}; \mathcal{A})$.

Alg. 2 summarizes the complete refinement procedure. Given anchors $\mathcal{A} = \{(t_j, l_j, u_j, w_j)\}_{j=1}^M$, we evaluate whether the trajectory satisfies each anchor interval within a local temporal window by a refinement loss:

$$\mathcal{L}_{\text{anch}}(\mathbf{x}; \mathcal{A}) = -\tau \sum_{j=1}^M w_j \text{LogSumExp}\left(-\frac{d(\mathbf{x}_{t_j-r:t_j+r}, [l_j, u_j])}{\tau}\right), \quad (13)$$

where $d(\cdot; [l_j, u_j]) = (\max(0, l_j - x) + \max(0, x - u_j))^2$ penalizes values outside the anchor band and r allows small temporal shifts.

Remark IV.1 (Connection to classifier guidance). Anchor Blended Sampling is an instance of classifier-guided diffusion [28]. Define the implicit “anchor classifier” $p(\mathcal{A} | \mathbf{x}^{(t)}) \propto \exp(-\mathcal{L}_{\text{anch}}(\mathbf{x}_\theta(\mathbf{x}^{(t)}), t; \mathcal{A}))$. The score of this classifier with respect to $\mathbf{x}^{(t)}$ is $\nabla_{\mathbf{x}^{(t)}} \log p(\mathcal{A} | \mathbf{x}^{(t)}) = -\nabla_{\mathbf{x}^{(t)}} \mathcal{L}_{\text{anch}} = g_t$, so the guided posterior mean $\mu_t + \gamma_t \Sigma_t g_t$ is exactly the classifier-guided update of [28] with guidance scale γ_t . Here, $\mathcal{L}_{\text{anch}}$ is an energy function, while $-\mathcal{L}_{\text{anch}}$ plays the role of the implicit log-classifier up to an additive constant. The blending step with $\mathbf{m}$ confines this guidance to anchor-relevant regions, leaving non-anchor timesteps undisturbed—an inpainting-style constraint analogous to [27] transposed from pixels to time.

Starting from a noised version $\mathbf{x}^{(T_e)}$ of the generated source $\mathbf{x}_\theta^{(0)}$, we run a short reverse editing process. At each timestep t , the posterior estimation $\tilde{\mathbf{x}}^{(t-1)}$ is updated using the negative gradient of the anchor loss:

$$p(\tilde{\mathbf{x}}^{(t-1)} | \mathbf{x}^{(t)}) = \mathcal{N}(\mu_t + \gamma_t \Sigma_t g_t, \Sigma_t), \\ g_t = -\nabla_{\mathbf{x}^{(t)}} \mathcal{L}_{\text{anch}}(\mathbf{x}_\theta(\mathbf{x}^{(t)}), t; \mathcal{A}).$$

Here, γ_t controls the guidance strength. To preserve non-anchor regions, we blend the guided latent with a noised version of $\mathbf{x}_\theta^{(0)}$:

$$\begin{aligned}\mathbf{x}^{(t-1)} &= \mathbf{m} \odot \tilde{\mathbf{x}}^{(t-1)} + (1 - \mathbf{m}) \odot \mathbf{x}_\theta^{(t-1)}, \\ \mathbf{x}_\theta^{(t-1)} &\sim \mathcal{N}(\sqrt{\bar{\alpha}_{t-1}} \mathbf{x}_\theta^{(0)}, (1 - \bar{\alpha}_{t-1}) \mathbf{I}),\end{aligned}$$

where $\mathbf{m}$ marks the anchor-relevant temporal regions. Thus, the base diffusion model generates a plausible trajectory, and the second energy-guided blended refinement locally corrects it according to the anchor constraints without retraining.

V. EXPERIMENTS

A. Experimental Settings

a) Dataset: We evaluate on the **Time-MMD** benchmark [5], a multi-domain multimodal dataset for time series analysis. **Time-MMD** contains paired numerical and textual data from real-world domains: **Economy**, **Energy**, **Security**, **Social Good**, and **Traffic**. The series are sampled at a weekly or monthly frequency and cover long temporal ranges from 1950 to 2024. Each numerical target is associated with aligned textual facts and reports, making the benchmark suitable for evaluating forecasting models that use both numerical histories and contextual text. Following the official **MM-TSFLib** protocol [5], we use the standard train/validation/test split with a ratio of 7:1:2.

b) Forecasting Setup: For monthly time series, we use a lookback length of $L = 36$ and prediction horizons $H \in \{6, 12, 18\}$. For weekly time series, we use $L = 96$ and horizons $H \in \{12, 24, 48\}$. For brevity, we report results averaged over all prediction horizons for each domain.

c) Baselines: We compare SCENARIODIFF with four groups of baselines. The first group includes numerical-only forecasters, covering Transformer-based models such as **Informer** [29], **FEDformer** [30], and **iTransformer** [31]. The second group includes LLM-prior methods, represented by **Time-LLM** [9] and **S²IP-LLM** [32]. The third group contains multimodal forecasting methods that use paired text, including **TaTS** [6] and **MM-TSF** [5]. The fourth group consists of probabilistic diffusion-based models such as **CSDI** [14] and **TMDM** [33]; for these models, we additionally report CRPS. We follow the **TSLib** [34] and **MM-TSFLib** [5] settings when available and tune hyperparameters on the validation set. Full baseline citations are given in Table I.

d) Metrics: We evaluate the deterministic forecasting accuracy using Mean Squared Error (MSE) and Mean Absolute Error (MAE) on standardized time series. For stochastic forecasting models, we additionally report the Continuous Ranked Probability Score (CRPS), which measures the quality of the predictive distribution. Lower values indicate better performance for all metrics.

e) Hyperparameters and Implementation Details: All models are implemented in PyTorch and trained on a single NVIDIA H100 GPU. Across the **Time-MMD** experiments, the diffusion forecaster uses $D = 6$ residual blocks, hidden dimension $d \in \{64, 128, 256\}$, and 8 attention heads. Models

are trained with Adam for 60–150 epochs, with batch size selected from $\{16, 32\}$, learning rate from $\{10^{-4}, 5 \times 10^{-4}\}$, and weight decay 10^{-6} . Forecast trajectories are generated using DDIM [35] with 50 sampling steps.

The Historical Context, Scenario, and Anchor Guidance Agents use Gemini 2.5 Flash. Agent outputs are generated offline and cached before training and inference. Text inputs are truncated to 512 tokens and encoded with a frozen **bert-base-uncased** encoder, yielding $d_{\text{emb}} = 768$ representations. To avoid temporal leakage, each forecast origin uses only documents and textual representations aligned with observed timesteps.

For Anchor Blended Sampling, we retain at most five anchor intervals, construct the edit mask with radius $r = 2$, set the soft-min temperature to $\tau = 0.10$, and perform $N_{\text{edit}} = 6$ guided DDIM reverse-editing steps. The anchor gradient is mask-restricted, clipped to unit norm, and applied with guidance scale $\gamma = 50$; non-anchor regions are blended with the noised source forecast.

B. Main Results

Table I reports average MSE and MAE over prediction horizons across the five **Time-MMD** domains. Overall, SCENARIODIFF achieves the strongest horizon-level performance, with the largest number of MSE/MAE wins across horizons and domains. The gains are most evident in event-driven domains, especially **Economy** and **Security**, while SCENARIODIFF also shows competitive performance on **Traffic**. These results suggest that noisy documents become more effective for forecasting when transformed into explicit scenario-level signals, rather than fused as unstructured text. Compared with numerical-only and LLM-prior baselines, SCENARIODIFF benefits from using LLM agents for contextual reasoning instead of direct numerical prediction. The agents extract historical evidence, generate scenario descriptions, and produce anchor points that guide a dedicated probabilistic forecaster. This separation of historical context, scenario-level guidance, and anchor-based refinement leads to stronger performance when text contains actionable event signals.

Table II reports CRPS for representative diffusion-based probabilistic forecasting models. SCENARIODIFF achieves the best event-domain average CRPS, indicating that hierarchical scenario guidance can also improve predictive distributions when textual evidence is informative. The improvement is not uniform across all domains, as strong numerical or diffusion-based baselines remain competitive in some settings. This domain-dependent behavior supports our main motivation: scenario-level guidance is most beneficial when external text provides actionable signals about future dynamics.

C. Ablation Studies

a) Contribution of Components: We evaluate the contribution of the three agent modules in SCENARIODIFF by removing one component at a time. Fig. 3 compares the full model with three variants: w/o Historical Context Agent (-His.), w/o Scenario Agent (-Scn.), and w/o Anchor Guidance

Table I: Overall results on **Time-MMD**. The best, runner-up, and third-best mean results are highlighted in **red**, **blue**, and **bold**, respectively. **Hor. 1st** counts the number of first-place results over all horizon-domain-metric entries, while **Ovr. 1st** counts first-place results after averaging over horizons within each domain and metric.

Models	Hor. 1st	Ovr. 1st	Economy		Energy		Security		Social Good		Traffic	
			MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
Informer [29]	0	0	0.891	0.774	0.456	0.525	127.556	6.595	0.973	0.599	0.248	0.405
Reformer [36]	0	0	1.036	0.853	0.676	0.632	122.835	6.285	1.046	0.647	0.292	0.443
Autoformer [37]	0	0	0.340	0.465	0.478	0.527	112.782	5.314	1.618	0.810	0.240	0.301
FEDformer [30]	2	0	0.286	0.410	0.394	0.457	113.725	5.382	1.238	0.666	0.231	0.271
PatchTST [38]	5	3	0.255	0.387	0.203	0.326	91.734	5.429	1.100	0.559	0.102	0.178
iTransformer [31]	5	2	0.280	0.398	0.227	0.344	113.248	5.432	1.235	0.568	0.208	0.238
PAttn [39]	3	0	0.237	0.376	0.267	0.388	83.117	4.956	1.196	0.569	0.104	0.181
DLinear [40]	0	0	0.579	0.636	0.391	0.448	106.504	4.665	1.524	0.935	0.284	0.415
FiLM [41]	0	0	0.460	0.556	0.375	0.469	108.179	5.158	1.608	0.949	0.236	0.326
TSMixer [42]	1	0	1.973	1.155	0.410	0.482	94.503	5.735	1.301	0.796	0.818	0.714
TiDE [43]	0	0	0.463	0.548	0.484	0.518	91.498	5.548	1.939	1.045	0.230	0.378
Time-LLM [9]	0	0	0.346	0.469	0.464	0.491	79.945	4.785	1.924	1.071	0.195	0.330
S ² IP-LLM [32]	4	0	0.273	0.417	0.224	0.343	76.184	4.381	1.025	0.594	0.191	0.310
TaTS [6]	7	3	0.924	0.732	0.457	0.540	124.457	6.368	0.886	0.541	0.184	0.312
MM-TSF [5]	0	0	0.816	0.737	0.397	0.481	127.460	6.573	0.959	0.580	0.229	0.382
CSDI [14]	4	3	1.396	0.943	0.545	0.531	96.372	6.092	0.908	0.491	0.124	0.256
TMDM [33]	3	1	1.038	0.765	0.320	0.398	76.270	4.177	1.305	0.673	0.103	0.187
NSDiff [44]	1	0	1.781	1.200	0.245	0.403	100.422	6.356	3.109	1.527	0.255	0.437
TimeDiff [15]	0	0	3.372	1.718	1.015	0.799	102.157	6.541	2.234	1.188	2.488	1.521
SCENARIODIFF	13	4	0.216	0.353	0.225	0.367	74.802	4.261	1.313	0.707	0.099	0.183

Table II: Average CRPS over prediction horizons for representative diffusion-based probabilistic forecasting models. We additionally report the average over event-driven domains, where textual scenarios provide actionable future signals.

Model	Econ.	Ener.	Sec.	Soc.	Traf.	Event Avg.
CSDI	0.231	0.138	0.805	0.109	0.043	0.392
TMDM	0.190	0.099	0.511	0.156	0.030	0.267
NSDiff	0.317	0.104	0.852	0.375	0.064	0.424
TimeDiff	0.436	0.197	0.883	0.266	0.251	0.506
SCENARIODIFF	0.094	0.112	0.584	0.189	0.036	0.263

Agent (-Anchor). Each bar reports the absolute MSE/MAE of the corresponding variant, so larger values indicate worse forecasting accuracy.

The results show that all three modules contribute to the final performance. Removing the Historical Context Agent causes the largest degradation in most domains, indicating that document-aligned historical evidence is important for grounding the forecast. Removing the Scenario Agent also hurts performance, especially in event-driven domains, confirming the value of forecast-horizon scenario descriptions. Removing the Anchor Guidance Agent leads to a smaller but consistent drop, suggesting that anchor guidance mainly acts as an inference-time refinement mechanism. Overall, the ablation results support the hierarchical design of SCENARIODIFF: historical summaries provide grounding, scenario descriptions provide future guidance, and anchor points provide localized refinement.

b) Event-driven Subset Analysis: To assess where scenario-level guidance contributes most, we partition the **Time-MMD** domains into event-driven domains (**Economy**,

Table III: Average MSE/MAE rank across event-driven and non-event-driven domains. Ranks are computed over all models in Table I for each reported MSE and MAE column, while this table reports representative strong baselines for compactness. Lower rank is better.

Model	Event-driven	Non-event-driven
TaTS	15.50	4.75
S ² IP-LLM	3.17	7.00
MM-TSF	14.17	8.25
PatchTST	4.17	3.25
iTransformer	7.00	6.75
PAttn	4.33	4.75
CSDI	15.33	3.50
TMDM	7.83	7.50
SCENARIODIFF	2.00	7.25

Energy, and **Security**) and non-event-driven domains (**Social Good** and **Traffic**). Table III reports the average rank over both MSE and MAE within each partition. SCENARIODIFF achieves the strongest mean rank on event-driven domains, outperforming representative numerical-only baselines such as PatchTST and PAttn, the LLM-prior method S²IP-LLM, and multimodal or diffusion-based baselines such as TaTS and TMDM. This result indicates that scenario-level guidance is most beneficial when external events and textual narratives provide actionable clues about future dynamics.

In non-event-driven domains, the advantage of SCENARIODIFF becomes less pronounced. Strong numerical-only, multimodal, and diffusion-based baselines achieve better average ranks, suggesting that these domains are more dominated by regular temporal patterns and that textual evidence provides fewer actionable signals. This contrast supports our central

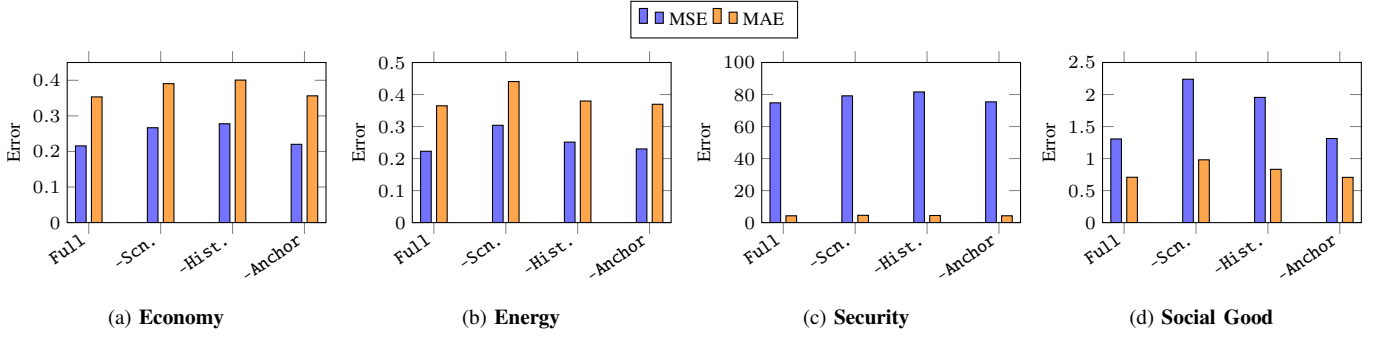


Fig. 3: Ablation study across four domains. Each subfigure reports the absolute MSE and MAE of the full SCENARIODIFF model and variants without the Scenario Agent, Historical Context Agent, and Anchor Guidance Agent.

Table IV: Quality of LLM-generated context in embedding space on **Energy**. *Coarse* uses history only; *Oracle* uses future (reference only). *Shuffle/Random* breaks alignment by shuffling contexts or sampling unrelated text.

Metric	Oracle vs. Coarse	Oracle vs. Shuffle/Random
Cosine similarity $\uparrow$	0.632	0.587 (Shuffle)
MMD $\downarrow$	0.376	0.579 (Random)
Fréchet distance $\downarrow$	0.432	1.532 (Random)

Table V: Local anchor diagnostic for sparse future anchors. MSE@r2 and MAE@r2 measure local alignment to each anchor band within radius $r = 2$. Gain MSE/MAE reports the endpoint improvement over a mean-value baseline that predicts the future as a mean trajectory.

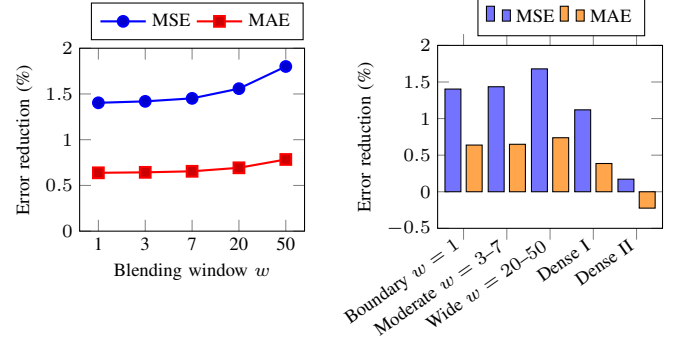
Dataset	Anchor NMSE@r2	Anchor NMAE@r2	Gain NMSE (%)	Gain NMAE (%)
Economy	0.121	0.229	47.63	37.50
Energy	0.393	0.427	-89.70	-24.19
Social Good	0.498	0.408	60.48	49.99
Traffic	0.040	0.100	59.58	39.96

motivation: scenario-level guidance is most useful when future dynamics are shaped by external events that are not fully recoverable from historical values alone.

c) *Quality of LLM-generated Context*: To verify whether the generated context carries meaningful signal beyond noise, we compare three text sources in SCS (see Eq. 9) on **Energy**: (i) *Coarse*, the history-only scenario generated by the Scenario Agent; (ii) *Oracle*, a reference scenario derived from ground-truth future values and used only for analysis; and (iii) misaligned baselines, *Shuffle/Random*, which break alignment by shuffling contexts across samples or replacing them with unrelated text. Table IV reports semantic and distributional agreement in the diagnostic SentenceTransformer [45] embedding space. The generated coarse scenarios are consistently closer to the oracle reference than the misaligned baselines across all metrics, indicating that the Scenario Agent produces signals distinguishable from random or misaligned text.

d) *Anchor Quality Analysis*: For each sparse anchor, we measure whether the predicted trajectory reaches the anchor band within a local radius $r = 2$. MSE@r2 and MAE@r2 measure local anchor alignment, while the gain metrics compare endpoint error against a flat mean-value baseline.

Table V shows that anchors can provide useful local signals.



(a) Sensitivity of anchor guidance to the blending window size. (b) Grouped parameter sensitivity of anchor guidance.

Fig. 4: Parameter sensitivity of anchor guidance on **Economy**.

Economy and **Traffic** have lower anchor errors and positive endpoint gains, suggesting that their anchors capture meaningful future deviations. In contrast, **Energy** has higher anchor error and negative gain, indicating that inaccurate values or temporal misalignment can misguide anchor-based refinement.

This diagnostic complements the full-horizon ablation: while the ablation evaluates the forecasting benefit of anchor guidance, this analysis examines whether the extracted anchors themselves are reliable. Overall, anchors are useful as sparse local constraints for abrupt future changes, but their effectiveness depends on anchor quality and temporal alignment.

e) *Parameter Sensitivity of Anchor Guidance*: We analyze the sensitivity of the anchor guidance module on **Economy** using parameter sweeps and report the mean MSE/MAE improvement over the no-anchor baseline. As shown in Fig. 4(a), anchor guidance is generally beneficial under stable settings. Moderate and wide blending windows provide more reliable improvements than very narrow windows, suggesting that anchor information should be incorporated smoothly into the diffusion trajectory rather than imposed as overly local constraints. The grouped results in Fig. 4(b) further show that increasing the number of anchors or applying stronger anchor expansion does not consistently improve performance. This suggests that aggressive anchor augmentation may introduce noisy or overly restrictive future constraints. Overall, anchor guidance works best with a compact anchor set and a sufficiently wide blending window.

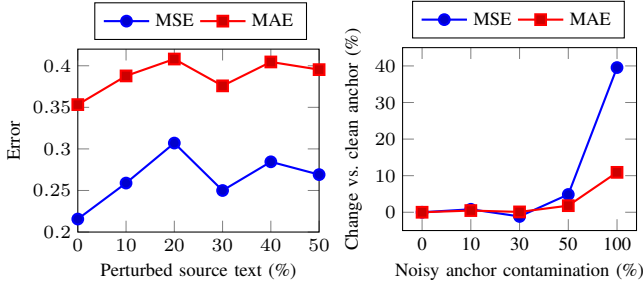


Fig. 5: Source-text perturbation sensitivity on **Economy**. Fig. 6: Robustness to noisy anchor contamination on **Economy**.

f) *Source-text perturbation*: To evaluate the sensitivity of SCENARIODIFF to corrupted textual signals, we perturb the source text at different ratios and report the mean forecasting errors over all prediction horizons. As shown in Fig. 5, the clean setting achieves the lowest MSE and MAE, indicating that textual context provides useful conditioning information. As the perturbation ratio increases, forecasting errors generally become larger, although the trend is not strictly monotonic. This suggests that SCENARIODIFF benefits from informative source text while retaining partial robustness to moderate textual noise.

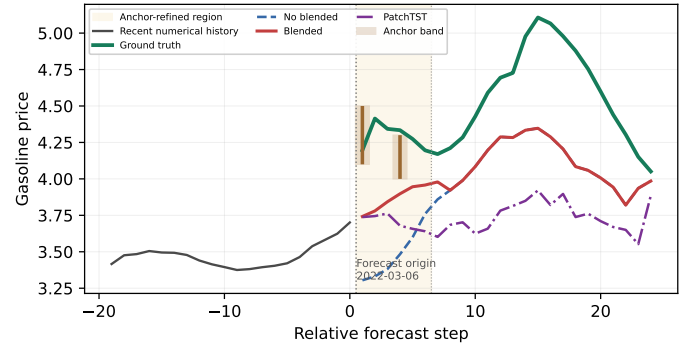
g) *Anchor Noise Sensitivity*: To evaluate robustness against erroneous guidance, we contaminate the anchor set with high-confidence noisy anchors sampled from the tails of each window’s historical distribution. In Fig. 6, moderate contamination has limited impact on forecasting accuracy, indicating that Anchor Blended Sampling can tolerate a small amount of noisy guidance. In contrast, severe contamination substantially degrades performance, especially when many noisy anchors are assigned high confidence. These results suggest that anchor guidance is robust to limited noise but remains sensitive to dense, high-confidence erroneous anchors.

D. Computational Cost

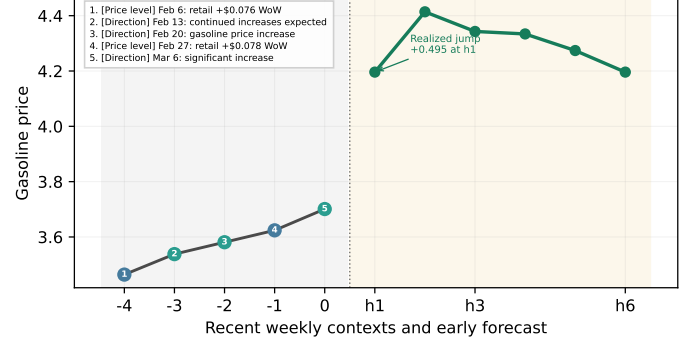
We separate offline LLM preprocessing from online forecasting. The three LLM agents are called once per input instance to generate and cache the stepwise context summaries, scenario description, and anchor points, with average latencies of 8–10 seconds, 13–15 seconds, and around 15 seconds, respectively. During online forecasting, SCENARIODIFF loads the cached agent outputs, encodes them with a frozen text encoder, and performs diffusion sampling. On a single NVIDIA H100 GPU with batch size 64, inference takes 0.9–1.3 seconds per batch using DDIM with 50 denoising steps. This cost is acceptable for weekly or monthly forecasting tasks, but SCENARIODIFF is not designed for strict real-time or high-frequency settings, where repeated LLM preprocessing or multi-trajectory diffusion sampling may become costly.

E. Case study

We present an event-driven case study from the **Energy** domain to illustrate how SCENARIODIFF uses contextual evidence. Fig. 7(a) compares the ground truth, the base diffusion forecast without blending, the anchor-refined forecast, and



(a) Mean forecasts, ground truth, and local anchor bands.



(b) Directionally aligned intrinsic evidence and the realized jump.

Fig. 7: Event-driven case in **Energy**. Historical evidence, scenario guidance, and anchor refinement jointly guide the forecast. Compared with PatchTST, blending improves the near-term trajectory and better follows the rising temporal pattern, although the shock magnitude remains underestimated.

the PatchTST baseline. While PatchTST stays close to the historical range and misses the upward transition, Anchor Blended Sampling refines the diffusion trajectory toward the local anchor bands and better follows the realized increase.

Fig. 7(b) shows the recent weekly observations and stepwise context summaries before the forecast origin. The summaries consistently indicate continued gasoline-price increases, providing directional evidence for an upward regime. The realized jump shortly after the forecast origin supports their role as historical grounding evidence.

Overall, the case study illustrates the three-level reasoning process of SCENARIODIFF: the Historical Context Agent extracts evidence of recent price increases, the Scenario Agent forms an upward forecast-horizon scenario, and the Anchor Guidance Agent provides local bands for event-relevant future steps. Although the full shock magnitude is still underestimated, the anchor-refined trajectory better captures the direction and timing of the event-driven change.

VI. CONCLUSION

We presented SCENARIODIFF, a hierarchical contextual reasoning framework for multimodal time series forecasting. By organizing contextual information into three levels—historical evidence (stepwise context summaries), future hypothesis (scenario description), and local constraints (anchor

points)—SCENARIODIFF conditions a Multimodal Diffusion Transformer on structured, interpretable signals rather than raw or implicitly fused text. Anchor Blended Sampling provides inference-time trajectory refinement grounded in classifier-guided diffusion. Experiments on **Time-MMD** show the greatest benefits in event-driven domains with actionable textual evidence. Future work will reduce dependence on external LLMs, improve scenario quality through structured generation, and evaluate the framework across additional multimodal forecasting benchmarks.

VII. ACKNOWLEDGEMENT

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APPENDIX

This appendix records the exact prompts used for the three LLM agents described in the methodology: the Historical Context Agent that extracts an intrinsic textual sequence aligned with the history, the Scenario Agent that produces a coarse natural-language description of the future horizon, and the Anchor Guidance Agent that generates anchor points for the sampling process. The prompts are designed so that the agents see only past information and aligned textual context, preventing temporal leakage from future values or documents.

A. Prompt for extracting intrinsic text (Historical Context Agent)

For each historical time step, the Historical Context Agent receives the aligned bag of text documents and is asked to summarize them into intrinsic (endogenous) dynamics and exogenous triggers in a compact JSON schema:

Prompt for Historical Context Agent

You are given a time step from a univariate time series.

TIME STEP

- sample_rate: {sample_rate}
- window_start: {step_start}
- window_end: {step_end}

{ds_block}CONTEXT ENTRIES (events/news/notes whose end_date falls inside this time step)
{ctx_block}

TASK

Classify the context for this single time step into two concise fields:

- 1) "intrinsic": expected, endogenous, or factual dynamics related to the series (e.g., regular seasonality, known policy schedule, realized facts within the step).
- 2) "trigger": unexpected or exogenous shocks that may have affected the series (e.g., surprise events, shocks, anomalies).

REQUIREMENTS

- Write short, specific phrases (less than 40 words each).
- If nothing notable, write "".
- Output EXACTLY this JSON (no markdown, no code block):
{"intrinsic": "<text>", "trigger": "<text>"}

The resulting "intrinsic" field is used as the history-aligned textual sequence $\text{intr}_{1:L}$; the "trigger" field is available for analysis or future extensions but does not directly enter the numeric forecaster in the current version.

B. Prompt for zero-shot coarse prediction (Scenario Agent)

The Scenario Agent receives only the lookback window and its stepwise intrinsic context, then produces a single-sentence coarse forecast for the future horizon. The prompt explicitly restricts the model to qualitative guidance and prohibits access to any future data, avoiding leakage:

Prompt for Scenario Agent

You are an expert time-series forecaster for the "{target_name}" in the domain "{domain_name}".

The data are sampled every {sample_rate}. You are given $L=\{\text{lookback_len}\}$ historical steps and must provide a high-level forecast for the next $H=\{\text{horizon_len}\}$ steps.

{ds_block}HISTORICAL VALUES (end_date \t value)
{lb_block}

STEPWISE CONTEXT (aligned to each end_date, derived from news/reports)
{it_block}

TASK

Provide a concise 1 sentence high-level forecast for the next {horizon_len} {sample_rate}s.
- Focus on directionality, volatility, and notable risks.

- Mention the most relevant drivers if apparent.
- Avoid numeric point forecasts; keep it qualitative.
- Avoid hedging language like "might" excessively.

OUTPUT

Return plain text only (no JSON, no bullets, no markdown).

STYLE RULES

- One sentence only (20-35 words).
- Do NOT narrate within-horizon sequence.
- Do NOT mention months/seasons or specific turning points.
- No numeric values; no exact dates.

The generated coarse description s_{LLM} is then encoded and injected into the diffusion forecaster via cross-attention on the future positions, as described in the main text.

C. Prompt for anchor-point guidance (Anchor Guidance Agent)

The Anchor Guidance Agent converts historical signals and contextual evidence into a sparse set of future anchor points for guided sampling. Given the lookback window, stepwise intrinsic context, and the coarse future summary generated from history only, it identifies potential abnormal regimes or sudden shocks and proposes uncertainty intervals at selected future steps. To prevent temporal leakage and over-constraint, the agent is instructed to output anchors only when credible evidence exists; otherwise, it returns an empty set.

Prompt for Anchor Guidance Agent

You are an expert time-series risk analyst. Your job is to propose a set of unexpected future value intervals for a diffusion-guidance rule, based on history value and mainly by context.

INDEXING CONVENTION

- $idx=0$ is the MOST RECENT observed point (end of lookback).
- Past points: $idx=-1, -2, \dots$
- Future points: $idx=+1, +2, \dots$
- Output uses t in $[1..H]$ and MUST correspond to $idx=+t$.

DE-SEASONALIZATION

- We removed seasonality using additive decomposition.
- $x_{ds} = x - \text{seasonal}$ (this equals trend + residual).
- Use x_{ds} to judge abnormality timing/strength, but output bands $[v_{lo}, v_{hi}]$ in ORIGINAL units.

TARGET

- series: {target_name}
- domain: {domain_name}
- values: ORIGINAL scale
- sampling frequency: every {freq} (one step = one {freq})
- history length: $L=\{L\}$
- horizon: $H=\{H\}$
- ORIGINAL stats: mean={ μ :.6g}, std={sd:.6g}, min={vmin:.6g}, max={vmax:.6g}
- x_{ds} stats: mean={d μ :.6g}, std={dsd:.6g}, min={dmin:.6g}, max={dmax:.6g}

RECENT HISTORY ($idx \backslash t_end_date \backslash t_orig_value \backslash tx_ds$) [last {show_last}]
{last_block}

CONTEXT ENTRIES (reports/search, within the lookback window)
{ctx_block}

TASK

Decide whether there is credible evidence of an "abnormal regime" or "sudden shock" in RECENT HISTORY and/or CONTEXT.

- If NO credible abnormal evidence exists, output an EMPTY list:
{{"points":[]}}.
- If YES, propose the MOST unexpected points in the next H steps, grounded

mainly in the CONTEXT ENTRIES.

CREDIBLE ABNORMAL EVIDENCE (generate points ONLY if at least ONE holds)

- 1) RECENT HISTORY shows a clear anomaly in x_{ds} : abrupt jump/drop, trend break, volatility spike, or unusually large magnitude relative to x_{ds} std.
- 2) CONTEXT explicitly mentions a concrete sudden event/driver likely to affect the series soon (timing implied).

If neither 1 or 2 holds, output `{{"points":[]}}`.

OUTPUT WHEN GENERATING POINTS

- Each point uses t in $[1..H]$ ($t=1$ is NEXT step after $idx=0$).
- Output $f=0$ and $type="NULL"$ for every point.
- Output an uncertainty interval $[v_{lo}, v_{hi}]$ in ORIGINAL units and confidence in $[0.1..1.0]$.

CALIBRATION RULES

- Always: $v_{lo} \leq v_{hi}$.
- Choose ONLY 1-4 points total. Choose t based on evidence timing.
- Bands must be plausible given ORIGINAL history:
 - Typical unexpected magnitude: $\sim 1-3 \times$ original std from recent mean/trend unless evidence supports extremes.
- Confidence $\Leftrightarrow$ band width:
 - Higher confidence leads to narrower band; lower confidence leads to wider band.

OUTPUT FORMAT (STRICT)

- Output MUST be EXACTLY one-line JSON.
- No markdown, no explanations, no extra keys.
- Schema:
`{{"points": [{{"t":int, "f":0, "type":"NULL", "v_lo":float, "v_hi":float, "confidence":float}}]}}`

Output JSON immediately (no leading text).

`{{"points": [...]}}`

We evaluate forecasting performance using mean absolute error (MAE), mean squared error (MSE), and Continuous Ranked Probability Score (CRPS). Let $\mathcal{I}$ denote the set of evaluated target positions across all test samples, prediction horizons, and variables after applying the evaluation mask. For each evaluated position $i \in \mathcal{I}$, let y_i be the ground-truth value and $\hat{y}_i$ be the point forecast. For probabilistic models that generate multiple forecast samples, we use the predictive mean as the point forecast when computing MAE and MSE.

a) *Mean Absolute Error*: MAE measures the average absolute deviation between the forecast and the ground truth:

$$\text{MAE} = \frac{1}{|\mathcal{I}|} \sum_{i \in \mathcal{I}} |\hat{y}_i - y_i|. \quad (14)$$

A lower MAE indicates that the predicted values are closer to the true values on average. In our deterministic evaluation, MAE is computed on the standardized time series following the official Time-MMD evaluation protocol.

b) *Mean Squared Error*: MSE measures the average squared deviation between the forecast and the ground truth:

$$\text{MSE} = \frac{1}{|\mathcal{I}|} \sum_{i \in \mathcal{I}} (\hat{y}_i - y_i)^2. \quad (15)$$

Compared with MAE, MSE penalizes large forecasting errors more strongly due to the squared term. As with MAE, MSE is computed on standardized values.

c) *Continuous Ranked Probability Score*: While MAE and MSE evaluate point forecasts, CRPS evaluates the quality of a predictive distribution. For a predictive cumulative distribution function F and observation y , the standard CRPS is defined as

$$\text{CRPS}(F, y) = \int_{-\infty}^{\infty} (F(z) - \mathbb{I}\{y \leq z\})^2 dz. \quad (16)$$

Equivalently, for a forecast distribution represented by samples $\{x^{(m)}\}_{m=1}^M$, CRPS can be estimated by

$$\text{CRPS} = \frac{1}{M} \sum_{m=1}^M |x^{(m)} - y| - \frac{1}{2M^2} \sum_{m=1}^M \sum_{n=1}^M |x^{(m)} - x^{(n)}|. \quad (17)$$

This score rewards forecast distributions that place high probability mass near the observed value while accounting for the dispersion of the generated samples.

In our implementation, we follow the commonly used quantile-based approximation of normalized CRPS. Given a set of quantile levels $\mathcal{Q} = \{0.05, 0.10, \dots, 0.95\}$, let $q_{\tau,i}$ denote the predicted τ -quantile at evaluated position i . The quantile loss is defined as

$$\text{QL}_{\tau}(y_i, q_{\tau,i}) = 2 [\tau(y_i - q_{\tau,i})\mathbb{I}\{y_i > q_{\tau,i}\} + (1 - \tau)(q_{\tau,i} - y_i)\mathbb{I}\{y_i \leq q_{\tau,i}\}]. \quad (18)$$

The reported normalized CRPS is then computed as

$$\text{nCRPS} = \frac{1}{|\mathcal{Q}|} \sum_{\tau \in \mathcal{Q}} \frac{\sum_{i \in \mathcal{I}} \text{QL}_{\tau}(y_i, q_{\tau,i})}{\sum_{i \in \mathcal{I}} |y_i|}. \quad (19)$$

Before computing nCRPS, both the target values and forecast samples are transformed back to the original data scale. The normalization by $\sum_{i \in \mathcal{I}} |y_i|$ makes the score scale-independent and comparable across datasets with different value ranges. Lower nCRPS indicates better calibrated and more accurate probabilistic forecasts.

This section provides additional quantitative results complementing the main tables in Chapter ?? . In particular, we include full MSE and MAE comparisons across all domains and horizons, using the standardized evaluation protocol described in Section ?? . The result is reported in Table VI, VII, VIII, and IX.

Table VI: MSE results in all horizons (Agriculture–Energy).

Domain	Agriculture				Climate		Economy			Energy		
Horizon	6	12	18	12	24	48	6	12	18	12	24	48
Informer	0.481	0.892	1.393	1.130	1.073	1.173	0.848	0.979	0.847	0.414	0.417	0.538
Reformer	0.323	0.933	1.573	1.168	1.098	1.165	0.766	0.943	1.400	0.571	0.688	0.770
DLinear	0.190	0.328	0.599	1.024	1.040	<u>1.044</u>	0.541	0.497	0.700	0.258	0.355	0.560
Autoformer	0.157	0.238	0.673	1.017	1.132	1.082	0.370	0.341	0.310	0.446	0.462	0.527
FEDformer	0.187	0.261	0.632	1.040	1.054	1.008	0.316	0.270	<u>0.273</u>	0.241	0.291	0.651
iTransformer	0.074	0.212	0.405	1.138	1.219	1.072	0.175	0.279	0.385	0.123	0.226	0.334
FiLM	0.138	0.262	0.553	1.218	<u>1.027</u>	1.058	0.436	0.589	0.355	0.253	0.363	0.509
Time-LLM	0.232	0.320	0.465	0.817	1.064	1.538	0.225	<u>0.246</u>	0.342	<u>0.117</u>	0.219	0.336
MCD-TSF	<u>0.110</u>	<u>0.236</u>	0.557	1.001	1.374	2.728	0.228	0.289	0.321	0.122	<u>0.188</u>	<u>0.306</u>
TaTS	0.158	0.251	0.534	1.006	1.418	1.311	0.690	0.603	1.478	0.355	0.488	0.528
MM-TSF	0.421	0.787	1.182	1.117	1.021	1.109	0.776	0.878	0.793	0.389	0.284	0.518
Ours	0.136	0.293	<u>0.445</u>	<u>0.885</u>	1.322	3.035	<u>0.211</u>	0.212	0.215	0.096	0.171	0.263

Table VII: MSE results in all horizons (Health–Traffic).

Domain	Health				Security		Social Good			Traffic		
Horizon	12	24	48	6	12	18	6	12	18	6	12	18
Informer	1.421	1.516	1.733	134.489	131.556	116.622	0.833	1.008	1.079	0.214	0.270	0.259
Reformer	<u>1.122</u>	<u>1.237</u>	<u>1.432</u>	127.034	128.065	113.407	0.904	0.991	1.243	0.284	0.265	0.326
DLinear	1.356	1.546	1.624	107.742	112.469	<u>99.300</u>	1.361	1.461	1.750	0.244	0.298	0.308
Autoformer	1.402	1.722	1.799	115.358	122.051	100.937	1.464	1.505	1.886	0.199	0.237	0.284
FEDformer	1.131	1.426	1.536	124.029	116.242	100.903	0.923	1.285	1.504	0.184	0.239	0.270
iTransformer	1.342	1.848	2.130	113.697	117.589	108.457	1.011	1.203	1.492	0.184	0.210	0.231
FiLM	1.840	1.667	2.151	<u>107.201</u>	<u>112.169</u>	105.168	1.318	1.900	1.605	0.191	0.277	0.241
Time-LLM	1.315	1.569	1.704	118.542	120.332	110.223	0.923	1.023	1.246	0.175	0.221	0.263
MCD-TSF	1.141	1.581	1.629	122.700	123.900	112.800	0.998	1.283	1.258	0.085	0.093	0.096
TaTS	1.033	1.183	1.252	125.885	131.938	115.548	<u>0.800</u>	0.915	0.943	0.167	0.192	0.194
MM-TSF	1.388	1.559	1.708	133.926	132.394	116.060	0.839	<u>0.969</u>	<u>1.069</u>	0.197	0.253	0.237
Ours	1.201	1.572	1.844	66.923	76.502	82.355	0.744	1.057	1.205	<u>0.087</u>	<u>0.097</u>	<u>0.113</u>

Table VIII: MAE results in all horizons (Agriculture–Energy).

Domain Horizon	Agriculture			Climate			Economy			Energy		
	6	12	18	12	24	48	6	12	18	12	24	48
Informer	0.582	0.775	0.910	0.845	0.820	0.850	0.748	0.826	0.749	0.484	0.500	0.589
Reformer	0.450	0.807	1.051	0.867	0.841	0.856	0.713	0.807	1.039	0.559	0.638	0.697
DLinear	0.344	0.428	0.537	0.807	0.812	<u>0.796</u>	0.619	0.579	0.709	0.368	0.429	0.547
Autoformer	0.320	0.395	0.654	0.783	0.831	0.816	0.488	0.469	0.438	0.498	0.508	0.574
FEDformer	0.372	0.421	0.635	0.818	0.822	0.785	0.440	0.394	<u>0.397</u>	0.365	0.395	0.609
iTransformer	0.187	0.367	<u>0.481</u>	0.878	0.906	0.825	0.313	0.401	0.479	0.253	0.340	0.439
FiLM	0.295	0.415	0.489	0.886	<u>0.800</u>	0.816	0.547	0.644	0.477	0.395	0.453	0.559
Time-LLM	0.358	0.433	0.494	0.702	0.826	0.975	0.368	<u>0.384</u>	0.437	<u>0.236</u>	0.346	0.432
MCD-TSF	<u>0.251</u>	<u>0.351</u>	0.564	0.835	0.957	1.296	0.366	0.408	0.442	0.265	<u>0.328</u>	<u>0.416</u>
TaTS	0.265	0.323	0.457	0.771	0.933	0.903	0.653	0.617	0.927	0.455	0.561	0.604
MM-TSF	0.539	0.718	0.856	0.839	0.798	0.823	0.708	0.779	0.724	0.477	0.392	0.575
Ours	0.278	0.463	0.520	<u>0.746</u>	0.939	1.342	<u>0.364</u>	0.368	0.347	0.234	0.321	0.395

Table IX: MAE results in all horizons (Health–Traffic).

Domain Horizon	Health			Security			Social Good			Traffic		
	12	24	48	6	12	18	6	12	18	6	12	18
Informer	0.832	0.808	0.868	6.680	6.651	6.454	0.504	0.623	0.670	0.372	0.435	0.407
Reformer	0.713	<u>0.757</u>	<u>0.823</u>	6.248	6.400	6.205	0.515	0.652	0.774	0.440	0.417	0.472
DLinear	0.773	0.833	0.876	<u>4.473</u>	<u>4.849</u>	<u>4.672</u>	0.858	0.895	1.053	0.381	0.429	0.437
Autoformer	0.830	0.985	0.966	5.169	5.794	4.978	0.741	0.765	0.923	0.280	0.287	0.337
FEDformer	0.750	0.851	0.897	5.876	5.369	4.899	0.514	0.707	0.776	0.238	0.277	0.297
iTransformer	0.765	0.925	1.064	5.297	5.531	5.469	<u>0.448</u>	0.560	0.694	<u>0.233</u>	0.223	0.257
FiLM	1.081	0.932	1.130	4.851	5.412	5.211	0.826	1.089	0.933	0.289	0.324	0.365
Time-LLM	0.773	0.825	0.958	5.514	5.659	5.472	0.503	0.605	0.675	0.256	0.283	0.305
MCD-TSF	<u>0.712</u>	0.814	0.903	5.300	5.550	5.350	0.549	0.620	0.680	0.152	0.162	0.171
TaTS	0.684	0.721	0.758	6.044	6.635	6.425	0.477	0.522	<u>0.624</u>	0.311	0.308	0.318
MM-TSF	0.818	0.818	0.873	6.626	6.691	6.403	0.455	0.613	0.672	0.352	0.416	0.378
Ours	0.737	0.819	0.965	3.925	4.174	4.250	0.384	<u>0.551</u>	0.611	0.152	<u>0.165</u>	<u>0.184</u>